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In [1]: import pandas as pd
import numpy as np
import joblib

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

from xgboost import XGBClassifier

RANDOM_STATE = 54895
```

```
In [7]: # Load data
df = pd.read_csv("../data/Crop_Yield_Fertilizer.csv")
```

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In [8]: df.head()
```

```
Out[8]:
```

	N	P	K	temperature	humidity	ph	rainfall	label	yield
0	90.0	42.0	43.0	20.879744	82.002744	6.502985	202.935536	rice	71.1994
1	85.0	58.0	41.0	21.770462	80.319644	7.038096	226.655537	rice	81.6201
2	60.0	55.0	44.0	23.004459	82.320763	7.840207	263.964248	rice	80.4731
3	74.0	35.0	40.0	26.491096	80.158363	6.980401	242.864034	rice	75.1781
4	78.0	42.0	42.0	20.130175	81.604873	7.628473	262.717340	rice	75.4855

```
In [9]: FEATURES = ['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']
TARGET = ['label']

X = df[FEATURES].copy()
y = df[TARGET].copy()
```

```
In [10]: label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)
```

```
C:\Users\preda\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\site-packages\sklearn\preprocessing\_label.py:110: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples, ), for example using ravel().
y = column_or_1d(y, warn=True)
```

```
In [11]: preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), FEATURES)
    ]
)
```

```
In [13]: model = XGBClassifier()
```

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n_estimators=600,
max_depth=5,
learning_rate=0.08,
subsample=0.9,
colsample_bytree=0.9,
objective='multi:softprob',
num_class=len(label_encoder.classes_),
eval_metric='mlogloss',
random_state=RANDOM_STATE
# n_jobs=-1
)

```

```

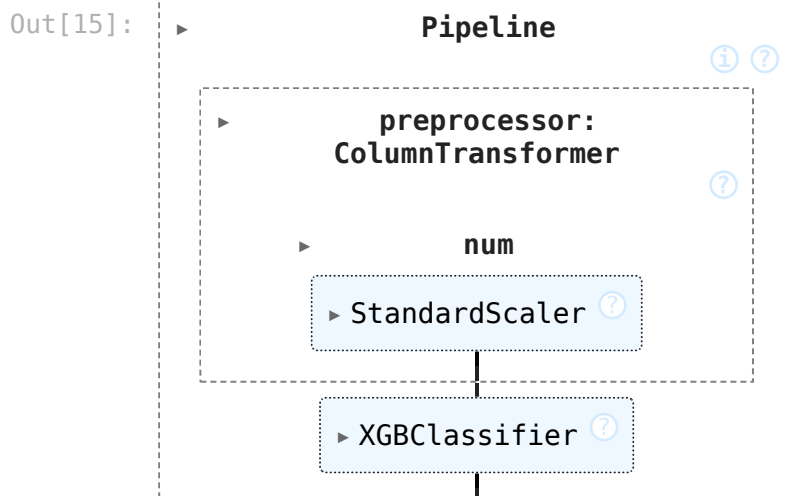
In [14]: pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('model', model)
])

```

```

In [15]: pipeline.fit(X, y_encoded)

```



```

In [17]: joblib.dump(pipeline, "../models/crop_model.pkl")
joblib.dump(label_encoder, "../models/crop_label_encoder.pkl")

print("Crop model trained and saved")

```

Crop model trained and saved